



Designing a PWM LED controller using 555 timer is absolutely straightforward and requires only a few components. From the diagram shown in Fig. 1 it is clear that the whole circuit is built around the 555 IC as the PWM generator. Hence, we can control the switching frequency and therefore the brightness of the LEDs with the help of passive and active components connected to it.

The 555 timer is a commonly used 8-pin integrated circuit (IC) that is used for a variety of applications, including pulse generator, oscillator, and timer. The operating voltage of the IC may vary from 4.5V to 16V. Fig. 2 shows different pins of the 555 IC.

PWM control using 555 IC

Before delving further into the circuit, let us understand the technique of PWM, which is also known as pulse duration modulation. It is a method of controlling the average power delivery by pulsating the DC current. In this circuit, the average power is controlled (reduced) by effectively chopping it up into discrete parts by connecting and disconnecting the power source in fast succession.

To achieve a PWM signal, the 555 timer is driven in an astable multivibrator mode. As the name suggests, the 555 timer has no stable state in this mode, but rather two quasi-stable states that change from high to low and low to high. In simpler words, in this mode the 555 timer turns on and off at a very high speed, which can be considered as 0 and 1 in the binary context, thus producing a square wave signal, as shown in Fig. 3.

Connecting the 555 IC

- Pin 1 is the ground pin and is connected to the -ve rail.
- Pin 8 is the VCC pin and is connected to the +ve rail.
- Pin 2 and pin 6 are the trigger pin and threshold pin, respec-

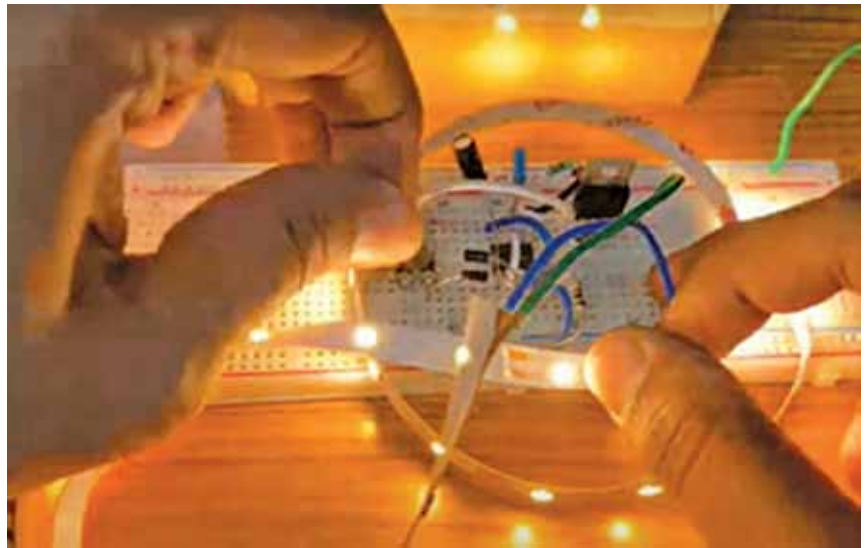


Fig. 5: Author's prototype showing maximum illumination of LED strip



Fig. 6: Brightness of LED strip reduced using potentiometer

tively, which are connected with the ground through a 1nF capacitor timing capacitor.

- Pin 3 is the output pin, which emits the PWM signal and is connected to the gate of the MOSFET.
- Pin 4 is the reset pin, which can be used to reset the operation. But we are not using it, so connecting it permanently to the +ve rail.
- Pin 5 is the control voltage pin, which is connected to the ground with a 100nF capacitor. It keeps the output stable and provides immunity against electrical noises.
- Pin 7 is the discharge pin, which is connected to the +ve input with a 1k resistor and is also connected to the inverted diode

setup and helps change the duty cycle.

IRFZ44N for high-speed switching

The 555 timer IC is capable of outputting a small amount of current as an output, which can be directly used only for a handful of LEDs. But in order to drive a long LED string, or control the speed of a motor, the current is not enough. To overcome this problem a MOSFET is used in the circuit whose gate is connected to the output of IC 555. The N-channel MOSFET used provides a negative switching control to the circuit.

For our circuit we are using N-channel MOSFET IRFZ44N as a